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SECTION 01

PROJECT 1

INDUSTRY INNOVATION AND INFRASTRUCTURE

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GROUP 9

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Problem

Self-checkout systems, which provide speed and convenience to both customers and companies, have become more common in retail settings in recent years. There is no denying that these systems offer benefits, but they also have a number of notable problems that may affect user satisfaction, operational effectiveness, and overall efficiency.

Figure shows long queue at the self-checkout system due to mistakes and requires the intervention of staff at AEON Taman Universiti, 25 April, 2024.

First of all, this self-checkout system may be prone to mistakes, especially **duplicate scans**. Due to either a broken scanner or their own oversight, customers may accidentally scan an item more than once. This means that he **needs to stop his current procedure and begin scanning each item one at a time**. Also, suppose after making his payment, he discovered there were a few scanning problems and **made two payments**. The customer might pay additional costs and experience inconvenience as a result of **having to seek help to correct the error and receive a refund for the duplicate item** (Garg, 2019).

Additionally, customers who make larger purchases could find it difficult to make sure that every item is accurately scanned before completing their transaction at the self-checkout because of the large number of items and the scanning area's physical space limitations. One common case is when customers mistakenly overlook items in their cart or trolley and **proceed to pay without realising they were not scanned.** This might happen due to distractions, multitasking, or plain human mistake. As a result, the unscanned items cause the store to lose out on income, which might result in long-term financial losses.

Figure shows an employee assisting a customer at the self-checkout lane during the grand opening of Fresh Thyme Market at City Foundry STL, November 10, 2021, in St. Louis, Missouri.

However, the most major issue contributing to scanning errors in self-checkout systems is the customer's unfamiliarity with the process. Many consumers **struggle to use the user interface (UI) and correctly scan products, which causes confusion and delays.** As a result, larger lines form at self-checkout terminals as customers hesitate (Wong & Hu, 2011), make mistakes, or seek assistance from staff. This not only reduces checkout efficiency, but also has an influence on the overall customer experience, producing aggravation for both consumers and employees.

Figure shows the available payment method at the self-checkout system at AEON BIG Falim

Furthermore, the self-checkout machine accepts credit cards and QR codes as payment methods, making it quick and easy for users to avoid handling cash. But there's a chance that **someone will take his credit card details and use them to make unauthorised payments**. For example, using "Visa payWave," which allows you to simply wave your card to pay, someone could steal the card and use it to make payments without your permission. This demonstrates that there are security issues with self-checkout systems and that strong safety precautions are required to protect payment information.

In summary, self-checkout systems provide benefits such as labour cost savings and convenience, but they also present problems. These include technical glitches, challenges for users and safety concerns. Therefore, self-checkout systems need to be improved in order to function more smoothly and efficiently handle these problems.

Solution

With acknowledgement of the current problem faced by the self check out counters, the "TakeNGo" technology shall be proposed for a better and hassle-free shopping experience. TakeNGo introduces a brand new method that requires neither cashier or customer themselves to carry out the process of checking out of the store, instead utilising in store mobile applications that relate to e-wallets for payment. Inside the store, they could simply take any items available, put it in a shopping bag or cart, and walk away effortlessly, needless to line up at counters or operate any system, maximising the efficiency while simultaneously providing the customer with a premium and personalised shopping experience.

How does TakeNGo manage this? First of all, all customers require a personal account of the TakeNGo app, and scan the QR code before entering, declaring that this customer is present in the store. When shopping, the items of their liking will be automatically added into a virtual cart inside of the mobile application right away, same goes to putting an item back, it will delete that item from the cart. This can be achieved by the security cameras that were placed all over the store which has live face recognition technology, and weighing features on the shelves, as well as implementing the concept of machine learning algorithms and artificial intelligence.

The take and go technology also saves many troubles such as packing groceries at check out and operating the self checkout system, while **effectively reducing the chances of error happening in the process**. If the customers are equipped with their own recycle bag, putting items inside of the bag does not affect the accuracy of the virtual cart. Therefore, the shopping process will be much more enjoyable without any troublesome experiences, since they can simply take the desired item and walk away.

With TakeNGo, there's **no more long queues** in front of counters. After selecting the preferable products, the customers can walk straight away from the store, and the total price of the items selected will be deducted from the balance of their e-wallet. Since the customers do not need to wait in turn for any operation like traditional counters or self checkout machines, there isn't any delay happening, hence queues are not an issue for the TakeNGo system.

Additionally, TakeNGo technology is private and secure for one and all customers that shop inside of the store. It uses biometric authentication face scanning for every customer that walks in the store, thus ensuring that only authorised individuals can make purchases, **significantly preventing unauthorised payments**. With TakeNGo, customers can enjoy a seamless and secure shopping experience, free from the worry of unauthorised transactions.

Furthermore, the points system and e-receipt are also available and completely applicable in this system. After leaving the store, customers can check receipts on the mobile application, which is perfect for personal accounting, expenses analysing and accounting, much easier than

using a paper receipt. Points accumulated can also be viewed on the app instead of only viewing it in store using the self checkout machine.

In the process of shopping, TakeNGo mobile application offers great features, including but not limited to the overview of the product placement, deals and offer sections, and personalised recommendation of products. The mapping section of the product allows customers to search for certain items by categories or brands, showing the precise location of the product. Deals and offer section shows customers the offer of the day, hence reducing the chances of loss of the store, while doing the same for personalised recommended products.

From the perspective of a store manager, TakeNGo technology is a great investment to have since it has eliminated the need for hiring cashiers, store assistants and security guards. It increases the safety of the store since a thief is almost impossible to happen unnoticeably, let alone robbery as the store was fully cashless.

Targeted Users

1. Busy Professional
   Targeting individuals with busy schedules who value efficiency and convenience can be very beneficial. Busy professionals often prioritise time-saving solutions that allow them to complete tasks quickly. TakeNGo provides a seamless shopping experience without queues or waiting in lines is a perfect fit for their lifestyle.

2. Inventory Manager
   TakeNGo can help the inventory managers to know the inventory status faster. When customers take items from the shelves, they can know about it in the shortest time possible. This is especially useful when there are only a few items left on the shelves; they can restock them immediately, ensuring that other customers won't face the frustration of not finding what they need.

3. Seniors and Disabled Persons
   Seniors often face the challenge of waiting in long lines due to limited mobility or discomfort, while disabled persons may have difficulty navigating crowded spaces. TakeNGo prioritises accessibility and convenience, allowing users to complete purchases efficiently and comfortably regardless of age or ability. Seniors and disabled persons can avoid these challenges by simply going to the supermarket, completing their purchases, and skipping the checkout line, allowing them to leave directly.

References

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